

Chapter 4

Beliefs, Mixed Strategies, and Expected Payoffs

To play a game well, a player has to think about what the other players are likely to do. This chapter makes that idea precise.

We will define a player's belief about her opponents' strategies, extend the idea of a strategy to allow for randomization, and see how to compute a payoff when the outcome is uncertain.

What Is a Belief?

In the Prisoners' Dilemma, player 1 might say to himself, "I think the other guy is likely to defect," or "I'm sure she'll cooperate." Statements like these are intuitive, but too vague to compute with.

Game theory makes them precise with a single tool: probability. Let p be the probability that player 1 thinks player 2 will cooperate, so $1 - p$ is the probability he assigns to defection. If $p = 1/2$, player 1 thinks the two actions are equally likely.

A belief need not be accurate. Player 1 can be confident that player 2 will cooperate and turn out to be wrong.

Notation for Beliefs

Definition. A **belief** of player i is a probability distribution over the strategies of the other players, written $\theta_{-i} \in \Delta S_{-i}$.

To be a genuine probability distribution, θ_{-i} must satisfy

$$\theta_{-i}(s_{-i}) \geq 0 \text{ for every } s_{-i} \in S_{-i}, \quad \sum_{s_{-i} \in S_{-i}} \theta_{-i}(s_{-i}) = 1.$$

In a two-player game this is simple: if player 2 has strategies $\{L, M, R\}$, player 1's belief is just three numbers, $\theta_2(L)$, $\theta_2(M)$, and $\theta_2(R)$, that are non-negative and sum to one.

Interpreting a Belief

There are two natural ways to read a belief, and game theory does not force us to choose between them.

- Player 2 might be **consciously randomizing**, for instance by flipping a coin to decide her action.
- Player 2 might instead be playing one fixed action, with player 1 simply **uncertain** about which one it is.

In a game with three or more players, a further subtlety appears: is it sensible for player 1 to think that players 2 and 3 randomize in a coordinated, correlated way? Nothing in the definition rules this out, so we allow it without requiring it.

Mixed and Pure Strategies

Definition. A **mixed strategy** for player i is the act of selecting her strategy according to a probability distribution, written $\sigma_i \in \Delta S_i$.

A mixed strategy and a belief are mathematically the same kind of object, a probability distribution over a strategy set. The only difference is whose strategies it describes: a belief is about the opponents, while a mixed strategy is a player's own plan.

Definition. A **pure strategy** is a mixed strategy that assigns probability one to a single strategy. Every pure strategy is therefore a special case of a mixed strategy.

Definition. A mixed strategy σ_i is **totally mixed** if it assigns strictly positive probability to every strategy in S_i : $\sigma_i(s_i) > 0$ for every $s_i \in S_i$.

Expected Payoff

Once player i has a belief about the others, or plans to randomize herself, she can no longer expect a single payoff for certain. We evaluate uncertain payoffs with their **expected value**: a weighted average, where each outcome's payoff is weighted by its probability.

Definition. Player i 's **expected payoff** from strategy s_i , given belief θ_{-i} , is

$$u_i(s_i, \theta_{-i}) = \sum_{s_{-i} \in S_{-i}} \theta_{-i}(s_{-i}) u_i(s_i, s_{-i}).$$

The same idea extends to a player evaluating her own mixed strategy, or to every player mixing at once: we simply weight each possible outcome by the probability that the players' independent randomizations produce it.

Example: Computing an Expected Payoff

	<i>L</i>	<i>M</i>	<i>R</i>
<i>U</i>	8, 1	0, 2	4, 0
<i>C</i>	3, 3	1, 2	0, 0
<i>D</i>	5, 0	2, 3	8, 1

Suppose player 1 believes player 2 plays *L* with probability $1/2$, *M* with probability $1/4$, and *R* with probability $1/4$: her belief is $\theta_2 = (1/2, 1/4, 1/4)$.

If player 1 chooses *U*, she gets 8 with probability $1/2$, 0 with probability $1/4$, and 4 with probability $1/4$, so

$$u_1(U, \theta_2) = \frac{1}{2}(8) + \frac{1}{4}(0) + \frac{1}{4}(4) = 5.$$

Discussion: Player 2's Expected Payoff

Using the same matrix, suppose instead that player 2 believes player 1 will select U with probability $1/2$, C with probability $1/4$, and D with probability $1/4$.

Player 2 plans to randomize herself, playing M and R each with probability $1/2$.

What is player 2's expected payoff?

Answer: Player 2's Expected Payoff

Six strategy profiles occur with positive probability. (U, M) and (U, R) each occur with probability $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$; the remaining four, (C, M) , (C, R) , (D, M) , (D, R) , each occur with probability $\frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$. These six probabilities sum to one, as they must.

Weighting player 2's payoff in each cell by its probability,

$$\frac{1}{4}(2) + \frac{1}{4}(0) + \frac{1}{8}(2) + \frac{1}{8}(0) + \frac{1}{8}(3) + \frac{1}{8}(1) = \frac{10}{8} = 1.25.$$

Payoffs Represent More Than Order

Until now, payoff numbers only needed to preserve the *order* of a player's preferences: any numbers that ranked outcomes the same way would do. Economists call this an **ordinal** ranking.

Once we allow uncertain outcomes, that is no longer enough: player 1 is exactly indifferent between U against $\theta_2 = (1/2, 1/4, 1/4)$ and D against L , both giving an expected payoff of 5.

If we changed the payoff of (D, L) from 5 to 4.5, player 1's ranking of the matrix's individual cells would not change at all, yet her preference between these two uncertain prospects would flip.

Expected-payoff comparisons require a **cardinal** scale, not merely an ordinal one: payoff numbers meaningful beyond their ranking, not just any numbers that preserve it.

Discussion: A CPBL At-Bat

A CPBL batter must commit to batting fast (f) or batting slow (s) before the pitcher's arm even starts forward, at the same moment the pitcher commits to throwing Fastball (F) or Slow Ball (S). The batter's payoff below is a 0–10 quality-of-contact score; the pitcher's is what's left from 10.

	F	S
f	8, 2	1, 9
s	2, 8	6, 4

Scouting reports say this pitcher throws Fastball with probability $3/5$ and Slow Ball with probability $2/5$: $\theta = (3/5, 2/5)$.

Compute the batter's expected payoff from batting fast, and from batting slow. Which should he choose?

Answer: A CPBL At-Bat

Batting fast,

$$u(f, \theta) = \frac{3}{5}(8) + \frac{2}{5}(1) = \frac{26}{5} = 5.2.$$

Batting slow,

$$u(s, \theta) = \frac{3}{5}(2) + \frac{2}{5}(6) = \frac{18}{5} = 3.6.$$

Against this belief, batting fast gives the higher expected payoff, even though batting slow correctly would have scored higher (6 vs. 8). The pitcher just doesn't throw it often enough to make it worthwhile.

Chapter in a Nutshell

- A belief is a probability distribution over the opponents' strategies; it need not be accurate, and can reflect either genuine randomization or mere uncertainty.
- A mixed strategy is a probability distribution over a player's own strategies; a pure strategy is the special case of probability one on a single strategy.
- Expected payoff weights each possible outcome by its probability, extending the payoff function to uncertain play.
- Once outcomes are uncertain, payoff numbers must respect preferences over probability distributions, not merely rank the certain outcomes.

Key Concepts

- Belief describes what a player thinks the others will do, formalized as a probability distribution θ_{-i} .
- Mixed strategy and pure strategy describe a player's own plan, with a pure strategy as the special, non-random case.
- Expected payoff is the probability-weighted average payoff a player anticipates under uncertainty.